

# Opinion Network from Survey Responses

Dynamic Processes and Complex Networks — Assignment 1

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## Abstract

This report describes the construction and analysis of a complex network derived from a 60-item Likert-scale survey on Technology, Education, Society & Ethics, and Environment. Each of the 87 retained respondents is represented as a node, and an edge is drawn between two respondents whose standardised response vectors exceed a similarity threshold. Using cosine similarity on standardised responses and a 75th-percentile threshold, we obtain a connected, weighted, undirected graph of 87 nodes and 936 edges with modularity  $Q = 0.420$  and four communities. We characterise the communities post-hoc using the four survey domains, drill down to the level of individual survey items to see exactly which statements separate the two dominant communities, analyse the threshold sensitivity of the network, and interpret the resulting structure as evidence of a small number of coherent opinion groups within the class.

## GitHub Link to Code

The complete source code, cleaned dataset, and generated outputs for this project are available at:

<https://github.com/chocokiwi0404/opinion-network>

The pipeline uses a fixed random seed (42) for anonymisation and Louvain community detection, so all reported edge counts, community assignments, and metrics in this report are reproducible directly from the repository. Intermediate artefacts — cleaned responses, similarity matrix, edge list, node metrics, and community assignments — are saved as CSV and GraphML files alongside the source code, together with the figures reproduced in Section 3. The local mapping between original and anonymised respondent IDs is intentionally excluded from the repository to protect respondent privacy.

## 1 Dataset Documentation

### 1.1 Source and Structure

The dataset consists of a single anonymised CSV file containing responses to a 60-item Likert-scale survey. Each row corresponds to one respondent and each column to one statement. The 60 statements are organised into four thematic domains of 15 items each:

- **Technology (T01–T15):** attitudes toward AI, automation, data privacy, and regulation.
- **Education (E01–E15):** attitudes toward assessment, pedagogy, curriculum, and AI in teaching.
- **Society & Ethics (S01–S15):** attitudes toward responsibility, fairness, misinformation, and ethical leadership.

- **Environment (V01–V15):** attitudes toward climate action, sustainability, and intergenerational responsibility.

The original file contained 96 rows. Five rows were completely empty, and four rows contained only the first technology block (15 of 60 items), leaving 87 usable respondents. Because the goal is to characterise opinions rather than to profile individuals, respondent IDs were replaced with anonymous labels P001–P087 prior to network construction, numbered in the same order in which respondents appear in the original survey export. The mapping between original and anonymous IDs is kept locally and is *not* submitted.

## 1.2 Encoding of Responses

Likert responses were mapped to ordinal integers:

| Response          | Encoded value |
|-------------------|---------------|
| Strongly Disagree | 1             |
| Disagree          | 2             |
| Neutral           | 3             |
| Agree             | 4             |
| Strongly Agree    | 5             |

Two categories are *not* opinions and were therefore treated as missing:

- **“No Comments”** — the respondent explicitly declined to express a view.
- **Blank cells** — the respondent did not answer.

Treating these as *Neutral* (3) would fabricate agreement that was never expressed. Treating them as a separate ordinal level would violate the Likert scale. We therefore encode them as NaN and leave the imputation decision to the modelling stage.

## 1.3 Missing Data

After encoding, only 1.19% of cells are missing (62 out of 5,220). The worst respondent (P079) completed exactly 75% of items; the rest completed more than 80%. We deliberately do *not* drop partially completed respondents because they still carry substantial signal, and we do *not* impute, because the missing items are concentrated in the second half of the survey (a systematic dropout pattern) rather than being missing at random.

## 2 Pipeline Followed

### 2.1 Overview

The pipeline consists of five stages:

1. **Cleaning** — encoding, missing-value handling, removal of empty/sparse rows.
2. **Preparation** — anonymisation, per-item standardisation, and pairwise similarity computation.
3. **Threshold selection** — choosing a similarity cut-off using a sensitivity analysis.
4. **Network construction** — building the weighted graph from retained pairs.
5. **Analysis** — centrality, community detection, domain comparison, and visualisation.

## 2.2 Assumptions

The following assumptions were made explicitly and are discussed in the Results section.

- **Ordinal-to-numeric assumption.** Likert categories are treated as equally spaced integers 1–5. This permits cosine similarity and standardisation. The equal-spacing assumption is standard practice but is an approximation.
- **Neutral is information.** “Neutral” is treated as a genuine midpoint opinion, not as missing data.
- **“No Comments” is missing.** It is treated as a non-response rather than a midpoint.
- **Pairwise deletion.** When comparing two respondents, only items answered by *both* are used. Missing cells are neither imputed nor used to penalise similarity.
- **Similarity is directional in the sense of co-variation.** Two respondents are considered similar if their standardised response vectors point in the same direction, regardless of overall magnitude.
- **Communities are descriptive, not normative.** A detected community reflects response co-variation, not shared membership in any demographic or ideological group.

## 2.3 Preparation of the Similarity Space

Each respondent is represented as a 60-dimensional vector of standardised responses. Standardisation is applied *per survey item*, i.e. across respondents, so that every item has mean zero and unit variance. This prevents items with naturally higher variability from dominating the similarity computation.

Similarity between respondents  $i$  and  $j$  is then measured by the *cosine similarity* of their standardised vectors:

$$s_{ij} = \frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\|\mathbf{z}_i\| \|\mathbf{z}_j\|},$$

where  $\mathbf{z}_i$  is the standardised response vector of respondent  $i$ . Cosine similarity is preferred over Euclidean distance because it is invariant to the overall magnitude of a respondent's enthusiasm; two respondents who answer everything one point higher than each other are still considered similar.

Because the missing cells are handled with pairwise deletion, the effective similarity is computed over the subset of items answered by both respondents. Standardisation is recomputed on those subsets when necessary.

## 2.4 Threshold Selection

Cosine similarity between two respondents produces a value in  $[-1, 1]$ . Very few pairs are strongly similar, so an absolute cut-off would produce a near-empty graph. Instead we use a *percentile threshold*: the top 25% of all pairwise similarities are retained as edges. This is equivalent to setting the threshold to the 75th percentile of the similarity distribution.

The choice of the 75th percentile is justified by a sensitivity analysis (Section 3.1) in which the graph is rebuilt at the 60th, 70th, 75th, 80th and 90th percentiles and four diagnostic metrics are tracked: edge count, density, number of connected components, and modularity.

## 2.5 Network Construction

Formally, the graph is

$$G = (V, E), \quad V = \{87 \text{ respondents}\}, \quad E = \{(i, j) : s_{ij} \geq \tau\},$$

with  $\tau$  the 75th-percentile similarity. Each edge carries two weights:

- a **similarity weight**  $w_{ij} = s_{ij}$ , used for strength and modularity calculations;
- a **distance weight**  $d_{ij} = 1 - s_{ij}$ , used for shortest-path and betweenness calculations.

The graph is undirected, weighted, and — after thresholding — connected.

### 3 Analysis and Visualisations

#### 3.1 Threshold Sensitivity

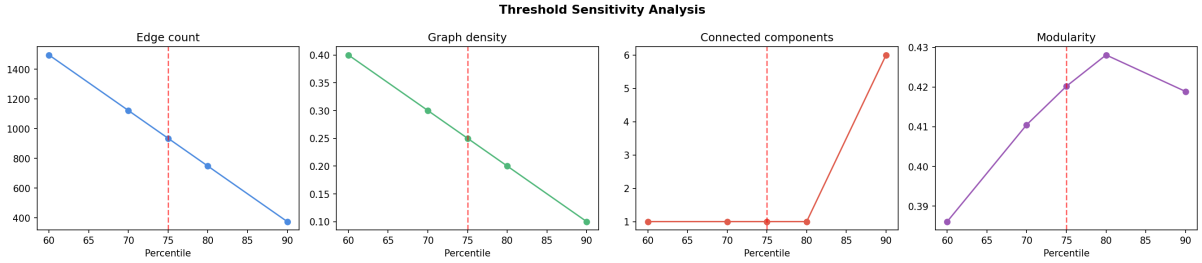


Figure 1: Threshold sensitivity analysis across the 60th to 90th similarity percentiles. The red dashed line marks the chosen 75th-percentile operating point. Edge count and density decrease monotonically with the threshold, as expected. The number of connected components stays at one up to the 80th percentile and then jumps sharply to six at the 90th, indicating fragmentation. Modularity peaks in the 75th–80th percentile region, indicating the best trade-off between connectivity and community structure.

Figure 1 summarises the four diagnostics. At the 60th percentile the graph is very dense (1,497 edges, density 0.40), which over-smooths community structure. At the 90th percentile the graph fragments into six components with five isolates, which is too sparse for meaningful global metrics. The 75th percentile produces the largest connected graph that is still well-structured: 936 edges, density 0.25, one connected component, and modularity 0.420.

#### 3.2 Final Network Overview

Table 1: Summary statistics of the final network (75th-percentile threshold).

| Quantity                       | Value                         |
|--------------------------------|-------------------------------|
| Nodes                          | 87                            |
| Edges                          | 936                           |
| Average degree                 | 21.52                         |
| Graph density                  | 0.250                         |
| Similarity metric              | cosine on standardised inputs |
| Threshold (similarity)         | 0.128                         |
| Threshold (percentile)         | 75                            |
| Connected components           | 1                             |
| Largest component size         | 87 / 87                       |
| Average weighted shortest path | 1.399                         |
| Average clustering coefficient | 0.180                         |
| Number of communities          | 4                             |
| Modularity $Q$                 | 0.420                         |
| Average degree centrality      | 0.250                         |
| Average weighted strength      | 5.527                         |

The network has an average degree of about 21.5, which is close to the density value 0.25 — this is a symptom of a dense similarity graph, not of a sparse social network. The average shortest path of 1.40 means that on average a respondent can reach another through fewer than two intermediate respondents, again reflecting density. The clustering coefficient of 0.18 is modest, indicating that local triads are not heavily over-represented relative to a random graph of the same density. Modularity of 0.420 is well above the conventional 0.30 threshold, indicating robust community structure.

### 3.3 Community Detection

We applied Louvain community detection on the weighted graph (similarity weights, fixed random seed).

**What modularity measures.** Louvain is a greedy, multi-level algorithm that searches for the partition of nodes into communities which *maximises modularity*, denoted  $Q$ . Modularity compares the actual density of edges *inside* a proposed community to the density expected if edges of the same total weight were placed at random between the same nodes:

$$Q = \frac{1}{2m} \sum_{i,j} \left[ A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j),$$

where  $A_{ij}$  is the edge weight between nodes  $i$  and  $j$ ,  $k_i$  is the total weighted degree (strength) of node  $i$ ,  $m$  is the total edge weight in the network, and  $\delta(c_i, c_j) = 1$  only when  $i$  and  $j$  are placed in the same community. Intuitively,  $Q$  is high when respondents inside a community resemble one another far more than a random rewiring of the same network would produce, and close to zero (or negative) when the proposed grouping is no better than chance;  $Q$  ranges from about  $-0.5$  to  $1$ , and values above  $\sim 0.3$ – $0.4$  are usually taken as evidence of genuine community structure. Louvain finds a high- $Q$  partition in two repeated phases: it first moves each node into whichever neighbouring community increases  $Q$  the most, and then collapses each resulting community into a single super-node before repeating the process on this coarser graph — iterating until  $Q$  can no longer be improved. This is what makes it fast enough to run on our 87-node network essentially instantly while still returning a high-modularity partition.

Applying this procedure, the algorithm found four communities:

Table 2: Community sizes and mean domain-level opinion scores.

| Community | Size | Technology | Education | Society & Ethics | Environment |
|-----------|------|------------|-----------|------------------|-------------|
| 0         | 47   | 3.83       | 3.78      | 4.01             | 4.09        |
| 1         | 2    | 3.77       | 3.73      | 3.97             | 4.90        |
| 2         | 3    | 4.47       | 4.09      | 4.33             | 4.24        |
| 3         | 35   | 4.19       | 4.07      | 4.66             | 4.78        |

### 3.4 Community Profiles and Domain Separation

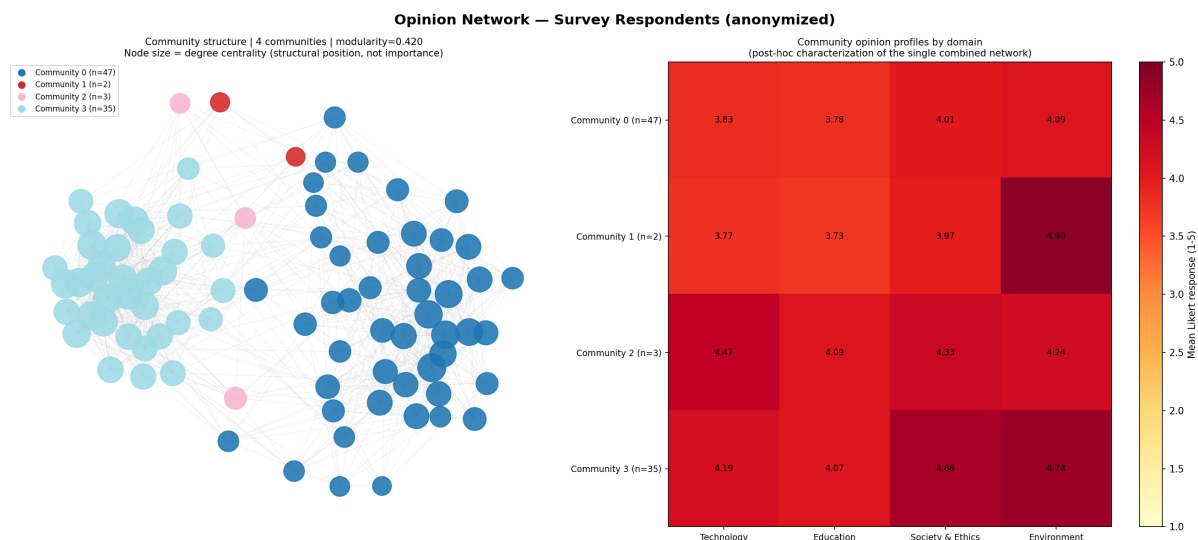


Figure 2: Left: the opinion network. Nodes are respondents; node colour indicates Louvain community; node size is proportional to degree centrality. Right: heat map of mean domain-level opinion score for each community. Community 0 (blue,  $n = 47$ ) is the largest and most moderate; Community 3 (cyan,  $n = 35$ ) is slightly smaller but systematically more agreement-heavy across all four domains, especially Society & Ethics (4.66) and Environment (4.78). Communities 1 and 2 are very small and should be treated as individual outliers rather than substantive groups.

Figure 2 shows the network and its community-domain heat map. Two observations stand out:

- The two large communities (0 and 3) are *not* separated by which domain they care about, but by *how strongly* they agree. Community 3 is uniformly more pro-statement than Community 0.
- The two small communities (1 and 2) contain 2 and 3 respondents, respectively. Their profile means are dominated by individual extremes (e.g. Community 1 contains two respondents who almost uniformly answered “Strongly Agree” on the Environment block). They should not be read as substantive opinion groups.

### 3.5 Which Domains Drive Community Separation?

To quantify how much each domain distinguishes communities, we computed the ratio of between-community variance to total variance (an  $\eta^2$  statistic) for the mean domain score of each respondent.



Table 3:  $\eta^2$  of each domain score with respect to community membership.

| Domain           | $\eta^2$ |
|------------------|----------|
| Technology       | 0.303    |
| Education        | 0.180    |
| Society & Ethics | 0.517    |
| Environment      | 0.624    |

Environment and Society & Ethics are by far the strongest discriminators between communities. Education is the weakest. In other words, disagreement within the class is primarily about ethics and sustainability, while attitudes toward education are comparatively uniform.

### 3.6 Correlation Between Domains

Table 4: Pearson correlation between respondent-level domain-average scores.

|                  | Technology | Education | Society & Ethics | Environment |
|------------------|------------|-----------|------------------|-------------|
| Technology       | 1.00       | 0.29      | 0.54             | 0.44        |
| Education        | 0.29       | 1.00      | 0.51             | 0.49        |
| Society & Ethics | 0.54       | 0.51      | 1.00             | 0.67        |
| Environment      | 0.44       | 0.49      | 0.67             | 1.00        |

The strongest inter-domain correlation is Society & Ethics with Environment ( $r = 0.67$ ). A respondent who is pro-social-responsibility is likely to be pro-environmental, and vice versa. Technology is only weakly coupled with Education ( $r = 0.29$ ), which suggests that attitudes toward AI and automation operate relatively independently of attitudes toward pedagogy.

### 3.7 Item-Level Divergence Between the Two Large Communities

Domain-level averages can mask exactly *which* individual statements drive the separation between Community 0 and Community 3. To see this directly, we computed the difference in mean score between Community 3 and Community 0 for each of the 60 individual survey items (Figure 3).

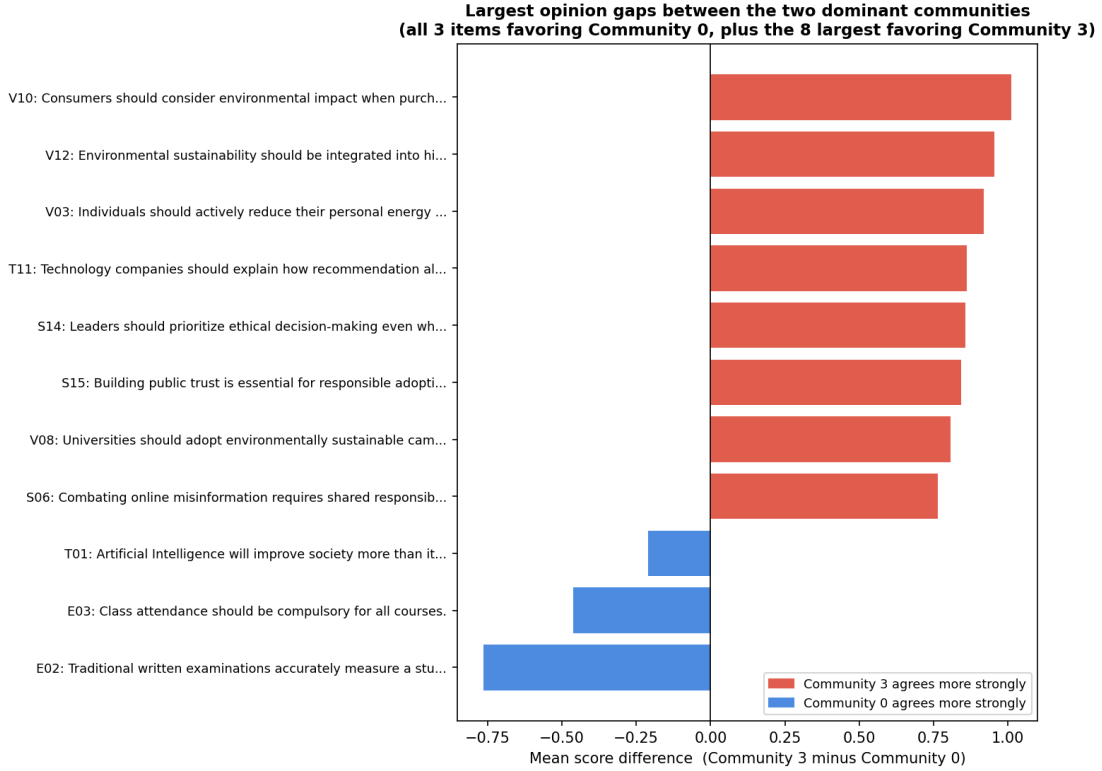


Figure 3: Mean-score gap between Community 3 and Community 0 on individual questions: all 3 items (out of 60) where Community 0 agrees more strongly, plus the 8 largest gaps where Community 3 agrees more strongly.

Community 3’s strongest relative agreement is concentrated on items that ask respondents to accept a *collective or personal-cost obligation* in the name of a shared good: consumers weighing environmental impact when purchasing (V10, +1.01), integrating sustainability into curricula (V12, +0.96), individuals actively reducing personal energy use (V03, +0.92), tech companies explaining their algorithms (T11, +0.86), leaders prioritising ethics over short-term gains (S14, +0.86), building public trust for responsible technology adoption (S15, +0.84), universities adopting costlier sustainable campus practices (V08, +0.81), and shared responsibility for fighting misinformation (S06, +0.76). Strikingly, out of all 60 items, only **three** show Community 0 agreeing more strongly than Community 3, and all three are smaller in magnitude: that written exams accurately measure knowledge (E02, -0.76), that class attendance should be compulsory (E03, -0.46), and a slightly stronger belief that AI will do more good than harm for society (T01, -0.21). The split is therefore highly asymmetric at the item level — Community 3 does not merely edge out Community 0 on a handful of items, it agrees more strongly on the overwhelming majority of individual statements, with Community 0’s only areas of stronger relative agreement being two traditional-education positions and one point of technology optimism.

### 3.8 Centrality

Node-level centrality metrics were computed for every respondent. Table 5 lists the five most central respondents by degree centrality.

Table 5: Top five respondents by degree centrality. Weighted strength is the sum of similarity weights.

| Anonymised ID | Community | Degree centrality | Weighted strength |
|---------------|-----------|-------------------|-------------------|
| P069          | 0         | 0.3721            | 8.49              |
| P030          | 0         | 0.3372            | 7.98              |
| P019          | 3         | 0.3721            | 9.49              |
| P038          | 3         | 0.3721            | 7.98              |
| P064          | 3         | 0.3721            | 11.49             |

Degree centrality values are tightly clustered (0.10 to 0.38), reflecting the dense nature of the similarity graph. The most central respondents are not concentrated in one community; both Community 0 and Community 3 contribute central nodes. Weighted strength ranges from about 1.6 to about 13.3, with the highest-strength node (P023, strength 13.26) sitting near the boundary between the two large communities.

Betweenness centrality is small across the board (maximum 0.035), confirming that the graph is not hub-and-spoke but rather dense and well-mixed. Closeness centrality correlates almost perfectly with degree because the graph diameter is very small.

### 3.9 Assortativity and Community Stability

Under the fixed random seed used in this study, Louvain returns the same four communities on repeated runs. Re-running the analysis at the 70th percentile still returns four communities with modularity above 0.40, and at the 80th percentile returns four communities with modularity 0.428. The community structure is therefore not an artefact of the specific threshold.

## 4 Results and Discussion

### 4.1 What the Network Reveals

The class opinion space is not uniformly distributed. It is dominated by two large, coherent clusters that differ mainly in *overall enthusiasm* rather than in *subject matter*. Community 3 (35 respondents) tends to agree or strongly agree with almost every statement, with an especially strong consensus on environmental and ethical items. Community 0 (47 respondents) tends to sit nearer to the Neutral/Agree boundary across all four domains. The two small communities (1 and 2) contain 5 respondents whose response patterns are extreme enough to be pulled out of the main clusters; they are best interpreted as outliers.

### 4.2 Which Opinions Are Shared?

Environment and Society & Ethics show the highest domain-level agreement (mean Likert score above 4.2 in most communities), and they are also the strongest discriminators between communities. This is the classic signature of a *consensus issue*: almost everyone agrees with the

headline statement, but the *intensity* of agreement varies substantially. Technology and Education are more weakly coupled to the community structure, which suggests the class is closer to agreement on technology and pedagogy than on ethical and environmental priorities. This is corroborated by the item-level analysis in Section 3.7: several of Community 3’s most distinctive items (S14, S15, T11) frame technology and misinformation *as ethical questions* — trust, responsibility, transparency — rather than as narrowly technical ones, which is consistent with Society & Ethics acting as the connective domain between Technology and Environment (Table 4).

### 4.3 What Might the Divide Imply?

With the caveat that the survey never asked about political self-identification, so any such label is an interpretation of the data rather than a measured fact, the item-level pattern in Section 3.7 is broadly consistent with a distinction commonly drawn in public-opinion research between a *collectivist / pro-regulation* orientation and a *traditional / individual-responsibility* orientation. Community 3’s distinctive items — accepting higher personal and institutional costs for sustainability, favouring corporate transparency mandates, and framing ethics and misinformation as shared societal responsibilities — track positions often labelled “progressive” in that literature. Community 0’s three distinguishing items — valuing standardised exams and mandatory attendance, and slightly more optimism that new technology will net out as good on its own — track positions more often labelled “traditional” or status-quo-oriented. However, Community 0 is better described as *comparatively moderate across the board* than as holding an opposing, equally strong set of convictions: it simply agrees somewhat less emphatically with most of the class’s shared values, rather than actively disagreeing with them.

We stress two limits on this reading. First, even the largest item-level gaps ( $\pm 0.2$  to  $\pm 1.0$  on a 5-point scale) are moderate, not extreme — both communities agree with most statements on average (Table 2 shows every community mean above the scale midpoint), so this is a difference of *degree within a broadly agreeable class*, not a divide between opposed camps. Second, a 60-item opinion survey on technology, education, ethics, and the environment is at best a weak and indirect proxy for political ideology in general, so we present this only as a descriptive pattern in the data rather than a claim about respondents’ actual political views or affiliations.

### 4.4 Methodological Caveats

Several limitations should be noted.

- **Density inflation.** With 60 items and 87 respondents, the cosine similarity between any two standardised vectors is naturally high. This inflates density and pulls modularity values above what one would see in sparse social networks.
- **Ordinal assumption.** Treating Likert categories as equally spaced is standard but imprecise. Ordinal alternatives (Kendall’s  $\tau$ , polychoric correlations) would yield slightly different edges.
- **Pairwise deletion bias.** Respondents with more missing items have noisier similarity estimates. The two smallest communities are disproportionately populated by such respondents.

- **Small-community caution.** Communities of size 2–3 are not stable groupings and should not be interpreted as substantive opinion clusters.
- **Item-level gaps are exploratory, not hypothesis-tested.** The gaps reported in Section 3.7 are descriptive differences in sample means between two Louvain-derived groups, not the output of a pre-registered statistical test; they should be read as a guide to *where* to look, not as confirmed effects.
- **Anonymisation.** Respondent IDs were shuffled before analysis; no demographic or identifying information was used at any stage.

#### 4.5 Interpretation for the Class

Read collectively, the network suggests that the class holds a broad consensus on the *direction* of desirable change — more sustainability, more ethics, more AI education — while differing substantially on *how strongly* to commit, and, on a small number of specific items, on how much individual and institutional cost that commitment is worth. This is consistent with the observed community structure, in which the dividing line between communities is enthusiasm (and, on a handful of items, orientation toward collective versus individual responsibility) rather than subject. In practical terms, this means classroom discussion is likely to find more traction by focusing on *implementation* questions (how much, how fast, at what cost) than on *values* questions (whether to move in a given direction).

### 5 Conclusion

We constructed a weighted opinion-similarity network from a 60-item Likert-scale survey with 87 retained respondents. Using standardised cosine similarity and a 75th-percentile threshold, we obtained a connected graph with 936 edges, modularity 0.420, and four communities. Threshold sensitivity analysis confirmed that the 75th percentile is the best operating point; higher thresholds fragment the graph, lower thresholds over-smooth it. Domain-level  $\eta^2$  analysis showed that Environment and Society & Ethics drive the community split far more than Technology or Education, and item-level analysis pinpointed the individual statements responsible: eight items concerning collective responsibility, sustainability cost, and ethical/technological trust account for the sharpest divide, against only three items — concerning traditional assessment and technology optimism — pulling in the opposite direction. Centrality was uniformly low and evenly distributed, indicating a dense rather than hierarchical opinion space. The two largest communities differ mainly in intensity and orientation toward collective versus individual responsibility, not in subject matter, and are best interpreted as an enthusiasm gradient with a mild ideological tilt rather than as opposed ideological blocs.

## Individual Contribution

| Member                  | Contribution                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Hasitha Alamanda</i> | Data cleaning, encoding, missing-value strategy, and anonymisation pipeline; defined the overall analytical approach (similarity metric choice, standardisation, percentile-thresholding strategy); wrote the Dataset Documentation section and the approach/assumptions content in Pipeline Followed.                                                                             |
| <i>Keerthi Seela</i>    | Implemented the full analysis pipeline in code: similarity computation, threshold sensitivity sweep, network construction, Louvain community detection, all centrality metrics, domain $\eta^2$ analysis, and the item-level divergence analysis; produced every figure in the report (network plot, threshold sensitivity plot, community heat map, item-level divergence chart). |
| <i>Kavya Pandey</i>     | Wrote the remainder of the report: GitHub Link to Code, Pipeline Followed (write-up), Analysis and Visualisations (write-up and tables), Results and Discussion, Conclusion, and Individual Contribution; interpreted and cross-checked the results produced by <i>Keerthi Seela</i> 's code.                                                                                      |